

Branching and workflows

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- This chapter adds **branches**, **tags**, **undo**, and **collaboration** patterns.

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- Default is usually **main**.
- Work on a side branch so **main** stays shippable while you try ideas.

Create a branch and switch to it

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- Dead end? Check out `main` again; delete or keep the branch as you prefer.

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- Find the tag later by name or hash when you need that snapshot.

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- Prefer **branches** for experiments so **main** rarely needs surgery.

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 7. **Delete** the merged branch.

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git checkout -b small-edits
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Example workflow

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2. Make changes, commit frequently

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git add my_new_file.py
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3. Push the branch

```
git push -u origin small-edits
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1. Sync with the remote and branch

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git checkout main  
git fetch --all --prune  
git rebase  
git checkout -b bugfix
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Another example workflow

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git commit -a -m "fix off-by-one in scoring"  
git push -u origin bugfix
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3. Open a **pull request** on GitHub; merge after review.

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- **Branch** for every task; keep **main** clean.
- **Write** descriptive commit messages (commit discipline).
- **Test** changes before committing.

Quick command reference

Command	Purpose
<code>git status</code>	Show working-tree state
<code>git add <path></code>	Stage files
<code>git commit -m "..."</code>	Record staged snapshot
<code>git log</code>	Browse history
<code>git diff</code>	Compare working tree to last commit
<code>git branch / git checkout -b</code>	List / create branches
<code>git push / git pull</code>	Sync with remote
<code>git fetch</code>	Download remote changes without merging
<code>git merge</code>	Combine branches
<code>git clone</code>	Copy a remote repo locally

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- Build the habit: **small commits, clear messages, frequent sync.**

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- Git tutorial · Git cheat sheet (PDF)

- [GitHub flow](#)
- [Git tutorial](#) · [Git cheat sheet \(PDF\)](#)
- [Ubuntu CLI cheat sheet \(PDF\)](#) · [Linux-fu](#)

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- The curious coder's guide to git